

Table 2. Empirical Comparison of Prime-Counting Identities at Powers of Ten: Correspondence Analysis between Traditional Error Terms ($\text{Li}(x)-\pi(x)$) and Proposed Composite Exclusion Terms (C_{comp}) for $10^1 \sim 10^{25}$

10^n	$\pi(10^n)$	$\pi(10^n)-4$	C_{comp}	$\text{Li}-\pi$
10^1	4	0	0	2
10^2	25	21	3	5
10^3	168	164	100	10
10^4	1,229	1,225	1,439	17
10^5	9,592	9,588	17,076	38
10^6	78,498	78,494	188,170	130
10^7	664,579	664,575	2,002,089	339
10^8	5,761,455	5,761,451	20,905,213	754
10^9	50,847,534	50,847,530	215,819,134	1,701
10^{10}	455,052,511	455,052,507	2,211,614,157	3,103
10^{11}	4,118,054,813	4,118,054,809	22,548,611,855	11,588
10^{12}	37,607,912,018	37,607,912,014	229,058,754,650	38,263
10^{13}	346,065,536,839	346,065,536,835	2,320,601,129,829	108,971
10^{14}	3,204,941,750,802	3,204,941,750,798	23,461,724,915,866	314,890
10^{15}	29,844,570,422,669	29,844,570,422,665	236,822,096,243,999	1,052,619
10^{16}	279,238,341,033,925	279,238,341,033,921	2,387,428,325,632,743	3,214,632
10^{17}	2,623,557,157,654,233	2,623,557,157,654,229	24,043,109,509,012,435	7,956,589
10^{18}	24,739,954,287,740,860	24,739,954,287,740,856	241,926,712,378,925,808	7,927,346
10^{19}	234,057,667,276,344,607	234,057,667,276,344,603	2,432,608,999,390,322,061	99,877,775
10^{20}	2,220,819,602,560,918,840	2,220,819,602,560,918,836	24,445,847,064,105,747,828	4,583,891
10^{21}	21,127,269,486,018,731,928	21,127,269,486,018,731,924	245,539,397,180,647,934,740	1,157,446,844
10^{22}	201,467,286,689,315,906,290	201,467,286,689,315,906,286	2,465,199,379,977,350,760,378	193,519,524
10^{23}	1,925,320,391,606,803,968,923	1,925,320,391,606,803,968,919	24,741,346,275,059,862,697,745	1,215,343,115
10^{24}	18,435,599,767,349,200,867,866	18,435,599,767,349,200,867,862	248,231,066,899,317,465,798,802	1,714,694,528
10^{25}	176,846,309,399,143,769,411,680	176,846,309,399,143,769,411,676	2,489,820,357,267,522,897,254,988	5,516,098,882